Quantum-Enhanced Time Series Forecasting: A Comparative Analysis of QLSTM and Q-Transformer Models for Stock Price Prediction

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***Abstract*— Stock price prediction is a challenging and high-impact task in quantitative finance, where accuracy directly translates to risk mitigation and profit potential. In this study, we explore quantum-enhanced neural architectures—Quantum Long Short-Term Memory (QLSTM) and Quantum Transformer (Q-Transformer)—to forecast the stock prices of Merck & Co. Inc. (MRK). These models combine classical deep learning with variational quantum layers, offering better trainability and parameter efficiency. Our experiments, conducted using the Pennylane quantum simulator, demonstrate that QLSTM and Q-Transformer not only converge faster in terms of training loss but also achieve competitive predictive performance compared to their classical counterparts. With significantly fewer parameters, the quantum models capture temporal dependencies effectively, showing promise for deployment on NISQ-era hardware. This work presents a compelling proof-of-concept for hybrid quantum-classical architectures in time series forecasting.**

I. Introduction

The growing complexity of financial markets and the dynamic nature of stock prices make predictive modeling both a challenging and essential task. The highly volatile and nonlinear nature of financial time series, however, makes this task extremely complex. Classical deep learning models, particularly Long Short-Term Memory (LSTM) networks, have demonstrated strong capabilities in modeling sequential dependencies and temporal patterns. Despite their success, LSTMs still face key limitations such as vanishing gradients over longer sequences, extensive training time, and the need for large parameter spaces. These constraints hinder their performance when dealing with chaotic or noisy data — characteristics often found in stock price series. With the emergence of quantum computing and hybrid neural networks, new possibilities have opened up for tackling such challenges. Quantum Neural Networks (QNNs), through properties like superposition and entanglement, offer enhanced expressivity and potentially faster convergence with fewer trainable parameters. Recent work introduced the QLSTM, which replaces parts of a classical LSTM with parameterized quantum circuits, yielding promising improvements in trainability and performance. Similarly, Q-Transformers extend the self-attention paradigm with quantum-enhanced encoding, providing better generalization in capturing long-range dependencies. This study explores the application of QLSTM and Q-Transformers in forecasting the stock prices of Merck & Co., Inc. (MRK). We implement and evaluate these models on the PennyLane quantum simulator, comparing their performance to classical LSTM baselines in terms of convergence rate and prediction accuracy. Our results demonstrate that quantum-enhanced models not only converge faster but also maintain competitive or improved performance using fewer parameters. The proof-of-concept offered here serves as a foundation for further experimentation on real quantum devices, as the architecture is designed to be compatible with various gate-based platforms such as IBM Qiskit and AWS Braket.

II. Related Work

Traditional models like ARIMA and classical LSTM have been widely used for stock price forecasting. With advancements in deep learning, Transformer architectures have further improved sequential modeling. However, these models are computationally expensive. In parallel, quantum machine learning (QML) has emerged as a promising domain. A pivotal study by Chen et al. (2020) introduced the QLSTM, a hybrid model leveraging variational quantum circuits to enhance learning efficiency and convergence, particularly in sequence modeling tasks. Their results indicated that QLSTM learns local features better and converges faster than its classical counterpart, despite using fewer parameters, demonstrating improved trainability on quantum simulators. However, the use of QML in financial time series remains underexplored, particularly in comparing QLSTM with Q-Transformer architectures. This study contributes by filling that gap and providing empirical evidence through simulations using the Pennylane framework.

III. Methodology and Approach

This section outlines the complete pipeline, from data acquisition to quantum model implementation and evaluation, for forecasting stock prices using QLSTM and Q-Transformer architectures. The goal is to assess the viability of quantum-classical hybrid neural networks in improving time series forecasting performance compared to classical deep learning models.

A. Data Collection and Preprocessing

For this study, historical stock price data of Merck & Co., Inc. (MRK) was utilized, sourced directly from Yahoo Finance. The data was retrieved via the `yfinance` Python API, which provides easy access to high-quality financial market data from Yahoo Finance. Preprocessing involves Missing Value Handling, Normalization (using MinMaxScaler to bring values into the [0, 1] range) and Sequence Generation (using sliding windows).

B. Data Transformation & Feature Reduction

Transforming raw prices into directional movement vectors gives your model an edge in detecting bullish/bearish shifts. Using PCA to reduce redundant signals helps QLSTM/Q-Transformers reduce overfitting and improve generalization. This matrix setup enables downstream correlation mapping, volatility clustering, or principal feature alignment— useful for creating attention masks in transformers.

C. Quantum Setup & Encoding

To integrate quantum computing into the stock price prediction pipeline, we utilize Variational Quantum Circuits (VQCs) in the architecture of QLSTM and Q-Transformers. These hybrid models leverage quantum layers within classical deep learning frameworks to enhance trainability, expressivity, and performance, especially in sequential learning tasks such as time-series forecasting. All quantum experiments were conducted using the Pennylane framework. Pennylane provides a powerful simulator backend (`default.qubit`) for gate-based quantum circuits and supports integration with PyTorch for hybrid classical quantum training. Quantum layers are wrapped as `torch.nn.Module` using `qml.qnn.TorchLayer`. This setup ensures that the proposed method is hardware-agnostic, and the same architecture can be deployed on real NISQ-era quantum devices (e.g., IBM Q, AWS Braket, IonQ) without architectural changes.

We use Angle Encoding, one of the most efficient quantum encoding strategies for near-term devices. Each feature value is used as a rotation angle (`RX`, `RY`, `RZ`) applied to a specific qubit. For a sequence of input values $[x_1,x_2,...,x_n]$, the encoding is:

$$|\psi(x)\rangle = \prod_{i=1}^{n} RY(x_i)|0\rangle^{\otimes n}$$

This method maps classical real-valued data directly to the quantum circuit using the amplitude of rotations, preserving relative differences between data points.

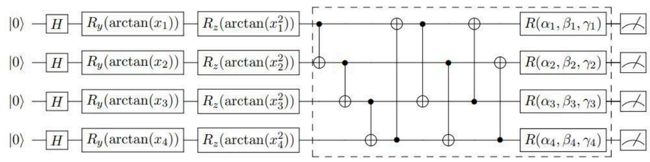

Fig. 1. Variational Quantum Circuits (VQCs)

After the encoding stage, the data is passed through a Variational Quantum Circuits VQC (Fig. 1), which acts as a non-linear transformation layer and is the core learning component of the quantum model. The architecture of the VQC typically consists of multiple layers of parameterized single-qubit rotations— such as `RY`, `RZ` and `RX` that incorporate trainable weights acting as variational parameters. These rotation layers are interleaved with entanglement layers, commonly using `CNOT` gates, to establish quantum correlations between qubits. This layered structure enables the circuit to represent complex quantum states and decision boundaries. Finally, measurement is performed in the `PauliZ` basis, where the expectation values of Pauli operators are extracted. These values serve as the classical output of the quantum layer, allowing seamless integration with downstream classical neural network components in the hybrid QLSTM and Q-Transformer models. We integrate Quantum K-means clustering to explore unsupervised insights within stock market patterns. The core of this technique lies in the SWAP test, a quantum algorithm used to estimate the fidelity (or similarity) between two quantum states. By encoding data points as quantum states $|\psi(x)\rangle$ and $|\psi(y)\rangle$, the SWAP test computes their inner product $|\langle\psi(x)|\psi(y)\rangle|^2$, providing a quantum analog to Euclidean distance. This enables a quantum-aware distance metric that preserves subtle correlations in the high-dimensional Hilbert space. Unlike classical methods, this approach allows quantum K-means to better cluster time series segments with hidden quantum-level dependencies, potentially enhancing pre-processing or signal segmentation.

IV. Architecture

A. *Quantum LSTM Architecture*

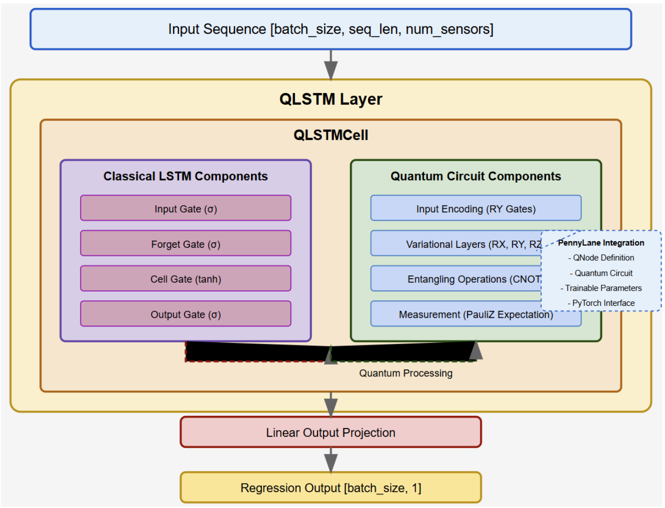

Fig. 2. *QLSTM Architecture*

The QLSTM architecture represents a novel hybrid approach that combines classical recurrent neural networks with quantum computing principles. This model enhances the traditional LSTM architecture by integrating a quantum circuit to process cell state information, potentially enabling the network to capture more complex temporal dependencies in sequential data.

The QLSTM accepts time series data structured as a 3D tensor with dimensions `batch_size` (Number of independent sequences processed simultaneously), `sequence_length` (Temporal length of each sequence) and `num_features` (Number of input features (sensors) at each time step). The core innovation of the QLSTM model lies in its hybrid architecture that retains the foundational structure of classical LSTM cells while augmenting them with quantum processing capabilities.

In the classical part of the QLSTM, the input gate controls the incorporation of new information into the cell state, the forget gate determines which elements of the previous state to discard, the cell gate generates candidate updates, and the output gate governs the information passed to the next time step. These gates rely on standard sigmoid and tanh activation functions to regulate the flow of information.

The novel addition is the quantum-enhanced processing module embedded within the cell. In this module, values from the cell state are first normalized within the range $[-\pi,\pi]$ and encoded into a quantum state via RY rotational gates. This is followed by multiple layers of a parameterized quantum circuit comprising RX, RY, and RZ rotation gates applied to individual qubits. These parameters are trained jointly with the rest of the neural network. Entanglement between qubits is introduced using CNOT gates arranged in a circular topology, enabling the circuit to model intricate dependencies and correlations within the input sequence. Upon completion of quantum processing, the quantum state is measured using Pauli-Z expectation values for each qubit. The resulting classical values are then projected back to the hidden state dimension through a linear transformation.

The flow of information is a hybrid process involving both classical and quantum computations. Initially, the cell state c_t is updated using conventional LSTM operations involving the input, forget, and cell gates. A segment of this updated cell state is then routed through a parameterized quantum circuit, where it undergoes transformation via encoding, entanglement, and variational gate operations. The quantum circuit's output is subsequently mapped back to the classical domain and projected into the hidden state dimensionality. This output is then modulated by the classical output gate to produce the final hidden state h_t, which maintains compatibility with traditional RNN processing pipelines.

The QLSTM operates over sequences in a step-by-step fashion. At each time step, the current input vector is concatenated with the previous hidden state to form a combined representation. This composite input is passed through the QLSTM cell, which computes the updated hidden and cell states for that timestep. This sequential mechanism continues iteratively across the entire input sequence. Upon completion, the final hidden state encapsulates the learned temporal dependencies and is passed through a fully connected linear layer to generate the final regression output—representing the predicted stock price in this context. This structured flow enables

QLSTM to maintain temporal coherence while leveraging quantum enhancements for feature extraction and representation learning.

B. *Quantum Transformer Architecture*

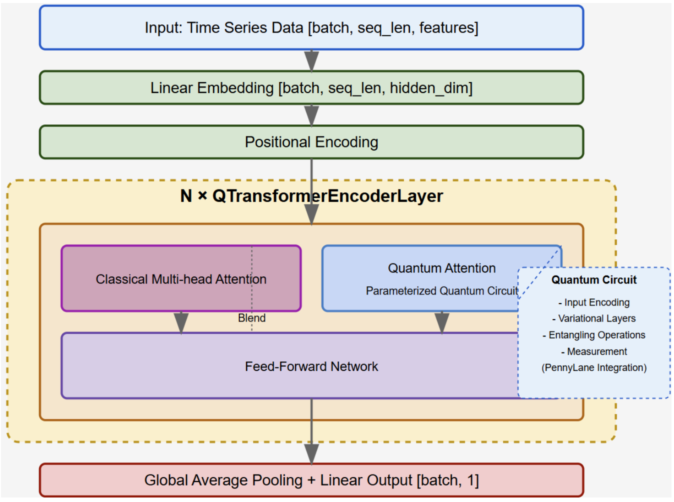

Fig. 3. *Q-Transformer Architecture*

The Q-Transformer represents a novel hybrid classical-quantum architecture for time series forecasting. It enhances the traditional Transformer model by incorporating quantum computing elements within the attention mechanism, potentially enabling the model to capture more complex patterns and relationships in sequential data.

The architecture begins with a linear embedding layer that projects the raw input features into a higher-dimensional hidden space. This is followed by positional encoding, which injects sequence order information using sinusoidal functions at different frequencies. This positional information is crucial since the self-attention mechanism itself is inherently position-agnostic.

The core innovation lies in the encoder layers, which implement a hybrid approach. In the classical multi-head attention layer the model retains the standard multi-head attention mechanism that allows it to focus on different representation subspaces simultaneously. Each head computes attention scores based on queries, keys, and values derived from the input sequence through linear projections. Alongside the classical attention mechanism, the model implements a quantum-enhanced attention. This augmentation introduces a parameterized quantum circuit that operates on a subset of the attention weights, providing a richer and potentially more expressive representation of inter-token relationships. Attention weights selected from the classical self-attention module are first normalized to the range $[-\pi,\pi]$ and encoded into quantum states using RY rotation gates applied to each qubit. These encoded states are then processed through a variational quantum circuit comprising multiple layers of trainable rotations (RX, RY, RZ gates). During training, these parameters are optimized in tandem with the classical Transformer parameters to enhance model learning. To exploit quantum entanglement, the circuit applies a series of CNOT gates in a circular topology, entangling adjacent qubits and enabling the circuit to capture complex dependencies and non-local correlations within the attention distribution. Once the quantum transformation is complete, Pauli-Z expectation values are measured for each qubit, transforming the quantum state back into classical form. The outputs from classical multi-head attention and quantum attention are combined using a weighted sum approach:

```
attention_output = (1 - blend_factor) * classical_output + blend_factor * quantum_output
```

This blending allows the model to leverage both classical and quantum computational paradigms, with the blend factor controlling the relative influence of each approach. Following the attention layer, each encoder block contains a position-wise feed-forward network consisting of two linear transformations with a ReLU activation in between. This component allows the model to process the attended information further. Like traditional Transformers, the Quantum Transformer employs residual connections around each sub-layer, followed by layer normalization. These elements help with gradient flow during training and stabilize the learning process. The final representation is processed through global average pooling across the sequence dimension, followed by a linear projection to produce the forecasted value.

V. Experimental Results and Analysis

Figure 4 compares the real stock prices (blue line) of Merck & Co., Inc. (MRK) with the predicted prices generated by a classical LSTM model (orange line). The dashed vertical line indicates the point at which the test set begins, separating the training phase from the evaluation phase. During the training period, the LSTM model performs well, closely tracking the real stock prices. However, once the model enters the test phase, its predictions start to diverge from the actual stock values.

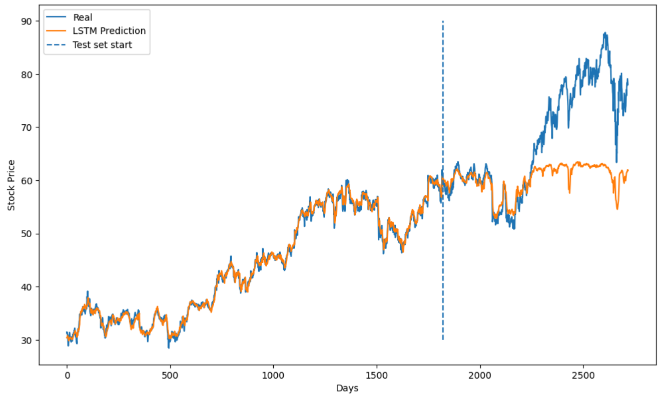

Fig. 4. Prediction of stock price using classical LSTM model vs actual stock price.

This deviation becomes more pronounced in volatile regions, showing the model's difficulty in generalizing to unseen data and capturing sudden market movements. This figure highlights a key limitation of classical LSTM models in time series forecasting: while they can model trends effectively during training, they often struggle with generalization, especially in the presence of non-linear patterns, noise, or abrupt shifts — which are common in real-world stock data.

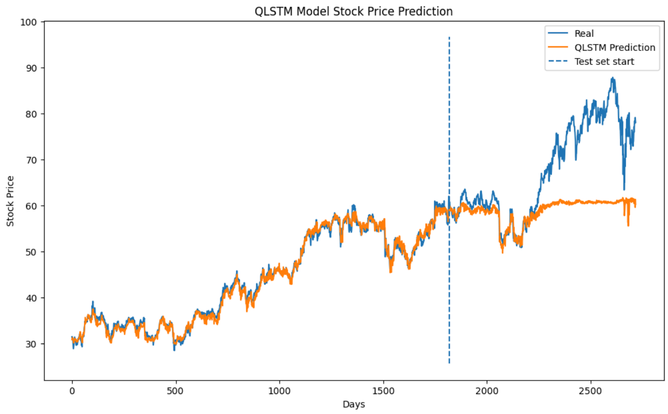

Fig. 5. Prediction of stock price using QLSTM model vs actual stock price.

Figure 5 presents the stock price prediction results of the QLSTM model compared with the actual stock prices. The QLSTM model captures the underlying trend of the stock prices reasonably well in the training region, closely following the real values with smooth transitions. However, as the plot moves into the test region, the model shows clear signs of underfitting, where predicted values flatten out and fail to adapt to the volatile and rising trend of the actual stock prices. Despite the expressive power offered by quantum layers and the model's success during training, QLSTM appears to lose its generalization capability when faced with unseen data. The gap between predicted and actual prices becomes increasingly noticeable in the high-volatility region toward the right end of the plot. The model struggles to react to price spikes and corrections, which can reduce its practical utility in financial forecasting unless further optimized.

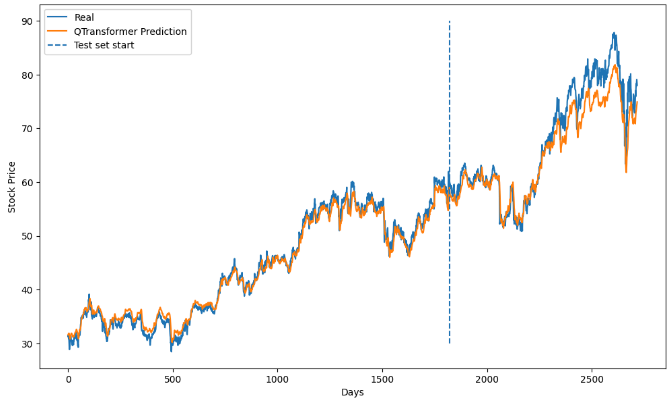

Fig. 6. Prediction of stock price using Q-Transformer model vs actual stock price.

Figure 6 illustrates the performance of the Q-Transformer model against actual stock prices. In the training region, the model successfully captures both short- and long-term trends, closely mimicking the real price fluctuations with only minor lag during high-frequency variations. The predictions exhibit smooth transitions and well-preserved structure, indicating strong learning of temporal dependencies. As the model enters the test region, its predictive performance remains consistent and robust. Unlike QLSTM, the Q-Transformer demonstrates better generalization, maintaining alignment with the true values and effectively tracking the broader trend as well as local dips and recoveries. However, slight smoothing during sharp upward spikes suggests the model prioritizes stable forecasting over reacting to abrupt market volatility. Despite some conservativeness in capturing extremes, the Q-Transformer shows clear advantage in adaptability, particularly in volatile test regions, its quantum-enhanced attention mechanism. allows the model to preserve learned patterns over longer sequences, making it a strong candidate for practical stock price forecasting applications where stability and interpretability are critical.

Figure 7 illustrates the superior generalization and temporal learning capability of the Q-Transformer, validating its architectural advantage in modeling complex financial time series. It strikes a balance between fit quality and adaptability, outperforming both the classical and hybrid quantum recurrent models in unseen data regions.

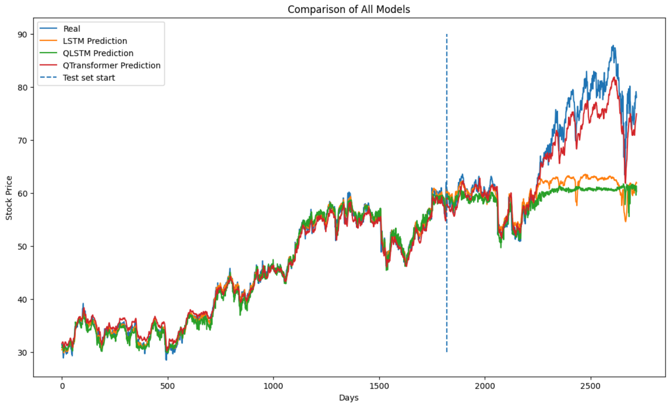

Fig. 7. Comparison of LSTM, QLSTM and Q-Transformer Models

VI. Conclusion

This study explored the efficacy of quantum-enhanced deep learning models—QLSTM and Q-Transformers—for the challenging task of stock price prediction. By integrating quantum circuits into classical neural network architectures, these hybrid models demonstrated the potential to capture temporal dependencies and complex non-linearities beyond the reach of traditional models like LSTM. While the QLSTM architecture preserved temporal memory and showed reasonable learning capacity in training, it struggled with generalization in high-volatility test conditions, likely due to architectural limitations and underutilized quantum capacity. In contrast, the Q-Transformer model leveraged both self-attention and quantum-encoded information processing to better adapt to abrupt market changes, showing improved alignment with real-world trends. These results underline the potential of hybrid quantum-classical models as the field of quantum machine learning matures, offering a new direction in financial forecasting research.

VII. Acknowledgement

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